

OS Unit 4 Question Bank — Memory Management & Virtual Memory

Operating Systems (OS) · Unit 4 · Memory Management & Virtual Memory · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. Define Contiguous Allocation. Derive or justify the key properties involved.
2. Define Paging. Derive or justify the key properties involved.
3. Discuss the role of Translation Lookaside Buffer in Memory Management & Virtual Memory.
4. Explain Segmentation with a neat diagram and a suitable example.
5. Explain Demand Paging with a neat diagram and a suitable example.
6. Discuss the role of Page Replacement Algorithms in Memory Management & Virtual Memory.

Part B — Long answer (10 marks)

1. Define Contiguous Allocation. Derive or justify the key properties involved.
2. Write short notes on Paging.
3. Illustrate Translation Lookaside Buffer with a worked example.
4. Illustrate Segmentation with a worked example.
5. Write short notes on Demand Paging.
6. Write short notes on Page Replacement Algorithms.
7. Discuss the role of Contiguous Allocation in Memory Management & Virtual Memory.
8. Illustrate Paging with a worked example.
9. Discuss the role of Translation Lookaside Buffer in Memory Management & Virtual Memory.
10. What is meant by Segmentation? Discuss its advantages and limitations.
11. Write short notes on Demand Paging.
12. Explain Page Replacement Algorithms with a neat diagram and a suitable example.